

PRE-LAB 4: Loops and Arrays

This pre-lab helps you practice **loops, arrays, and basic functions** in C. The problems are simple and inspired by **everyday scenarios**. Each question includes a solution.

Question 1: Average Steps

Description:

A person recorded the number of steps they walked for 5 days. Write a function that calculates the **average steps**.

Function Template:

```
int average_steps(int steps[], int size);
```

Example: Input: {8000, 9000, 10000, 7000, 8500} Output: 8900

Solution:

```
#include <stdio.h>

int average_steps(int steps[], int size) {
    int sum = 0;
    for(int i = 0; i < size; i++) {
        sum += steps[i];
    }
    return sum / size;
}

int main() {
    int steps[] = {8000, 9000, 10000, 7000, 8500};
    int avg = average_steps(steps, 5);
    printf("Average steps: %d\n", avg);
    return 0;
}
```

Question 2: Weekly Temperature Chart

Description: You have weekly temperature data stored in an array. Write a function to print a temperature chart using *, where each * represents 1 degree.

Function Template:

```
void print_temperature_chart(int temps[], int size);
```

Example: Input: {3, 5, 2} Output:

```
Day 1: ***  
Day 2: *****  
Day 3: **
```

Solution:

```
#include <stdio.h>  
  
void print_temperature_chart(int temps[], int size) {  
    for(int i = 0; i < size; i++) {  
        printf("Day %d: ", i + 1);  
        for(int j = 0; j < temps[i]; j++) {  
            printf("*");  
        }  
        printf("\n");  
    }  
}  
  
int main() {  
    int temps[] = {3, 5, 2};  
    print_temperature_chart(temps, 3);  
    return 0;  
}
```